WORKED SOLUTION

STEP III 1993, Question 2

Pure • Inflection and Polar Area

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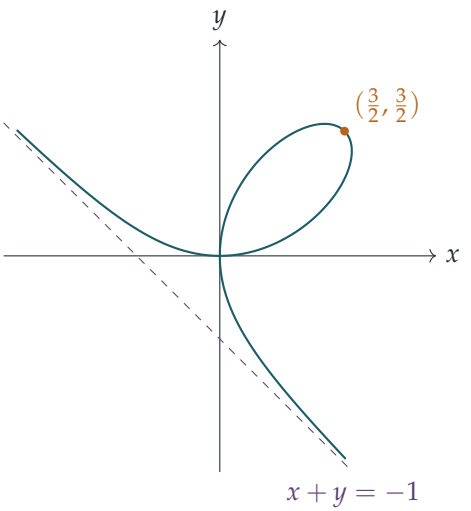
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q2

The key idea

At a point of inflection the second derivative must vanish (and change sign), so the plan for (i) is to find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ by implicit differentiation and see when it can be zero. The algebra looks heavy, but the equation of the curve itself collapses the numerator to something tiny. For (ii), the cubic terms and the xy term have different degrees, so in polar coordinates a factor r^2 cancels and r comes out as an explicit function of θ.

Solution



The sketch shows the curve (the folium of Descartes): a loop in the first quadrant and two branches in the second and fourth quadrants approaching the asymptote $x + y = -1$.

Part (i)

Differentiating $x^3 + y^3 = 3xy$ implicitly with respect to x,

$$3x^2 + 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} = 3y + 3x\frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{N}{D}, \quad N = y - x^2,\ D = y^2 - x,$$

wherever $D \neq 0$. Differentiating the quotient, and using $\frac{\mathrm{d}N}{\mathrm{d}x} = y' - 2x$ and $\frac{\mathrm{d}D}{\mathrm{d}x} = 2yy' - 1$ with $y' = N/D$,

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= \frac{(y' - 2x)D - N(2yy' - 1)}{D^2} = \frac{(N - 2xD) - (2yN^2/D - N)}{D^2} \\ &= \frac{2(ND - xD^2 - yN^2)}{D^3}.\end{aligned}$$

Now expand the bracket:

$$\begin{aligned}ND &= x^3 + y^3 - x^2y^2 - xy, \\ xD^2 &= xy^4 - 2x^2y^2 + x^3, \\ yN^2 &= y^3 - 2x^2y^2 + x^4y,\end{aligned}$$

so

$$ND - xD^2 - yN^2 = 3x^2y^2 - xy - xy^4 - x^4y = -xy\,(1 + x^3 + y^3 - 3xy).$$

On C we have $x^3 + y^3 - 3xy = 0$, so the bracket is just 1 and

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{-2xy}{(y^2 - x)^3}.$$

Ask yourself Where could this be zero, and what about points where the formula breaks down?

The second derivative vanishes only if $x = 0$ or $y = 0$. On C, $x = 0$ forces $y^3 = 0$, and $y = 0$ forces $x^3 = 0$; either way we are at the origin, which we may assume is not a point of inflection. Away from the origin and from points with $y^2 = x$, the second derivative is continuous and non-zero, so it cannot change sign: no inflection there.

That leaves the points with $D = y^2 - x = 0$, where y is not a differentiable function of x. If N were also zero there, we would have $x = y^2$ and $y = x^2$, so $x = x^4$, giving $(0,0)$ or $(1,1)$; but $(1,1)$ is not on C since $1 + 1 \neq 3$. So at every such point other than the origin $N \neq 0$: the tangent is vertical, and we should treat x as a function of y. The equation of C is symmetric in x and y, so swapping the roles of the letters in the work above gives

$$\frac{\mathrm{d}^2x}{\mathrm{d}y^2} = \frac{-2xy}{(x^2 - y)^3},$$

which is defined (because $x^2 - y = -N \neq 0$) and non-zero (because $xy \neq 0$ away from the origin). So the curve does not change its sense of bending at these points either.

Hence C has no point of inflection.

Part (ii)

Put $x = r\cos\theta$, $y = r\sin\theta$. Then

$$r^3(\cos^3\theta + \sin^3\theta) = 3r^2\cos\theta\sin\theta,$$

and for $r \neq 0$ we can divide by r^2:

$$r = \frac{3\cos\theta\sin\theta}{\cos^3\theta + \sin^3\theta}.$$

For $0 \leq \theta \leq \frac{\pi}{2}$ the numerator and denominator are both non-negative (and the denominator is positive), so $r \geq 0$, with $r = 0$ exactly at $\theta = 0$ and $\theta = \frac{\pi}{2}$. So as θ runs from 0 to $\frac{\pi}{2}$ the point goes once round the loop, starting and finishing at the origin. The area is

$$A = \frac{1}{2}\int_0^{\pi/2} r^2\,\mathrm{d}\theta = \frac{9}{2}\int_0^{\pi/2} \frac{\cos^2\theta\sin^2\theta}{(\cos^3\theta + \sin^3\theta)^2}\,\mathrm{d}\theta.$$

Dividing top and bottom by $\cos^6\theta$ (positive for $0 \leq \theta < \frac{\pi}{2}$),

$$A = \frac{9}{2}\int_0^{\pi/2} \frac{\tan^2\theta\sec^2\theta}{(1+\tan^3\theta)^2}\,\mathrm{d}\theta.$$

Substituting $u = \tan\theta$, $\mathrm{d}u = \sec^2\theta\,\mathrm{d}\theta$, with u running from 0 to ∞,

$$A = \frac{9}{2}\int_0^{\infty} \frac{u^2}{(1+u^3)^2}\,\mathrm{d}u = \frac{9}{2}\left[-\frac{1}{3(1+u^3)}\right]_0^{\infty} = \frac{9}{2}\cdot\frac{1}{3} = \frac{3}{2}.$$

The improper integral converges because the antiderivative tends to 0 as $u \to \infty$. So the area of the loop is $\frac{3}{2}$.

Checks

- The loop's tip is where $y = x$: $2x^3 = 3x^2$ gives $(\frac{3}{2}, \frac{3}{2})$, at distance $\frac{3}{\sqrt{2}} \approx 2.12$ from the origin. A loop of that length and width roughly 1 should have area of order 1 to 2, which fits.
- Using the parametrisation $x = \frac{3t}{1+t^3}$, $y = \frac{3t^2}{1+t^3}$ (the line $y = tx$ meets C there), we compared $\frac{-2xy}{(y^2-x)^3}$ with the parametric second derivative at random points, and checked that the curvature never changes sign on any branch.
- Midpoint quadrature of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ and a shoelace sum round the parametrised loop both give 1.500000.

Common slips

- Expanding the second derivative fully and never noticing that $x^3 + y^3 - 3xy$ appears inside it; the simplification is the whole point.
- Forgetting the points with vertical tangents, where $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is undefined rather than zero.
- Writing $\frac{1}{2}\int r\,\mathrm{d}\theta$ instead of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ for the area.

Beyond STEP

Implicit differentiation is justified by the implicit function theorem: near any point where $\partial F/\partial y \neq 0$, the equation $F(x, y) = 0$ really does define y as a smooth function of x. The origin, where both partial derivatives vanish, is a singular point where two branches cross, which is why it needed special treatment. The folium is also a rational curve, parametrised by lines $y = tx$ through its double point, an idea that leads into algebraic geometry. For the analysis behind all this, see Bartle and Sherbert, *Introduction to Real Analysis*.

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